

$$\left| \begin{array}{l} y^2 = x^2 + ax + b \checkmark \\ \text{char}(F) > 3 \end{array} \right.$$

$$E = \{ (x, y) \in \mathbb{Z}_2^2 \mid \underbrace{y^2 + xy = x^3 + 1} \} \cup \{\infty\}$$

$$= \{ (0, 1), (1, 0), (1, 1), \infty \}$$

$$\left| \begin{array}{l} \cancel{2xy} + xy' + x'y + y = 3x^2 \\ m = \left. \frac{y'}{y} \right|_{(0,1)} = \frac{3x^2 - y}{2y + x} \right|_{(0,1)} \\ = \frac{-1}{2} = -\frac{1}{2} \end{array} \right.$$

$$P = (0, 1), \quad 2P = \infty$$

$$Q = (1, 0), \quad 2Q = (\quad , \quad)$$

$$\left| \begin{array}{l} y' \Big|_{(1,0)} = \frac{3}{1} \\ = 1 \quad (\text{in } \mathbb{Z}_2) \end{array} \right.$$

$$x_3 = m^2 - 2x_1$$

$$y - 0 = (x - 1)$$

$$y = x - 1$$

$$\underline{y^2 + xy = x^3 + 1}$$

$$(x-1)^2 + x(x-1) = x^3 + 1$$

$$\Rightarrow \cancel{x^2} + 1 + \cancel{x^2} + x = x^3 + 1$$

$$\Rightarrow x^3 - x = 0$$

$$\Rightarrow x(x^2 - 1) = 0$$

$$\Rightarrow x(x-1)^2 = 0$$

$$\underline{x_3 = 0 \quad \text{as} \quad x_1 = x_2 = 1}$$

$$Q = (1, 0)$$

$$2Q = (0, 1) = P$$

$$\& \quad 2P = \infty$$

$$\underline{y_3 = 0 - 1 = 1}$$

$\therefore 4Q = \infty$ & so that

the order Q is 4. since $|E| = 4$

$$E \cong C_4$$

$$\bullet E = \{ (x, y) \in \mathbb{F}_4 \times \mathbb{F}_4 : y^2 + y = x^3 + 1 \} \cup \{ \infty \}$$

Recall $\mathbb{F}_4 = \{0, 1, \alpha, \alpha^2\}$ where $\alpha^2 = 1 + \alpha$
(hence $\alpha^3 = 1$)

$$E = \{ (0, 0), (1, 0), (0, 1), (1, 1), \infty \}$$

if $x = \alpha$ $y^2 + y = \alpha^3 + 1 = 1 + 1 = 0$
 $\alpha \in \mathbb{F}_4$

$$\begin{aligned} y^2 + y &= 0 \\ y(y+1) &= 0 \\ y &= 0 \text{ or } 1 \end{aligned}$$

if $x = \alpha^2 \neq 1 + \alpha$, $y^2 + y = (\alpha^2)^3 + 1 = 1 + 1 = 0$

✓ $(\alpha, 0), (\alpha, 1), (\alpha^2, 0), (\alpha^2, 1)$

$\in E$

Find other points & for a point $P \in E$
Find the order of P

• Do the same thing for $y^2 + y = x^3$ over $\mathbb{F}_4$

$$\bullet E = \{ (x, y) \in \mathbb{F}_4 \times \mathbb{F}_4 : y^2 + xy = x^3 + 1 \} \cup \{ \infty \}$$

Show that $E = \{ (0, 1), (1, 0), (1, 1), (\alpha, 0), (\alpha, \alpha),$
 $(\alpha^2, 0), (\alpha^2, \alpha^2), \infty \}$

where $\mathbb{F}_4 = \{0, 1, \alpha, \alpha^2 = 1 + \alpha\}$

Find the order of $Q = (0, 1)$ & $P = (1, 0)$
[show $4P = Q$ & $2Q = \infty$]

Since $|E| = 8$, $E = \langle P \rangle$ so that $\underline{E \cong C_8}$

Application of Elliptic curves.

1. Factorization of integers.

$n = 589$

Consider the curve $y^2 = x^3 + 4x + 9$
 $E(\mathbb{Z}_n) = \{(x, y) \in \mathbb{Z}_n^2 \mid y^2 = x^3 + 4x + 9\} \cup \{\infty\}$

Set $P_1 = (2, 5) \in E$.

$P_2 = 2P_1 = (x_2, y_2)$,

$\left. \begin{aligned} x_2 &= m^2 - 2x_1 \\ y_2 &= m(x_1 - x_2) - y_1 \\ m &= \frac{3x_1^2 + a}{2y_1} \end{aligned} \right\} \text{ formulae.}$

$m = \frac{16}{10}$

$10^{-1} \equiv 59$ so that $m = 355 \in \mathbb{Z}_{589}$

Then $P_2 = (x_2, y_2) = (564, 156) \in \mathbb{Z}_{589}^2$

$3P_1 = (x_3, y_3) = (148, 48) \neq \infty$

$4P_1 = (303, 572)$

$5P_1 = (\quad , \quad) \neq \infty$ $6P_1 = (\quad , \quad) \neq \infty$

$7P_1 = \underline{3P_1 + 4P_1}$

$m = \frac{524}{155}$

To find 155^{-1} we perform.

$\left\{ \begin{aligned} x_7 &= m^2 - x_3 - x_4 \\ y_7 &= m(x_3 - x_4) - y_3 \\ m &= \frac{y_4 - y_3}{x_4 - x_3} \end{aligned} \right.$

extended division algorithm

$$589 = 3 \times 155 + 124$$

$$155 = 1 \times 124 + 31$$

$$124 = 4 \times 31 + 0$$

$$\text{GCD}(589, 155) = \underline{31}$$
$$589 = 31 \times 19$$

So 589 is not a prime.

Pollard's (p-1) algorithm (Recall)

$$\underline{d_n} = \text{GCD}(2^{n!} - 1, m), \quad n \in \mathbb{N}$$

$$\underline{d_n \mid d_{n+1} ?} \quad \text{yes} \quad \underline{(2^{n!} - 1) \mid (2^{(n+1)!} - 1) ?}$$

$$\left[\text{same } (a-1) \mid (a^m - 1) \right] \parallel \{ (2^{n!})^{n+1} - 1 \}$$

$$d_n = (2^{n!} - 1, m)$$

Ex)

let $\underline{m = 403}$

$$\checkmark d_1 = \text{GCD}(2^{1!} - 1, 403) = 1$$

$$d_2 = \text{GCD}(2^{2!} - 1, 403) = \text{GCD}(3, 403) = 1$$

$$d_3 = \text{GCD}(2^{3!} - 1, 403) = \text{GCD}(64, 403) = 1$$

$$d_4 = \text{GCD}(2^{4!} - 1, 403) = \text{GCD}(2^{24} - 1, 403) = 13 \quad \checkmark$$

$$\underline{16777215} \quad \underline{13 \mid 403}$$

$$d_5 = \text{GCD}(\underline{16777215} - 403)$$

$$\text{GCD}(a, b) = \text{GCD}(c, b)$$

$$\text{if } \underline{a \equiv c \pmod{b}}$$

not required
because
you have
a divisor
already

Ex Next $m=589$ ✓

Find $d_1, d_2, \dots$

$$d_i = 1 \quad \underline{1 \leq i \leq 4}$$

Q $d_5 = ?$